

## KHANTY-MANSIYSK, Russian Federation

WMO: 239330

Lat: 61.0282N

Lon: 69.1179E

Elev: 76

StdP: 14.66

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -36.2      | -31.2 | -41.7                         | 0.5 | -35.9 | -36.5 | 0.7 | -31.0 | 17.8                  | 7.7  | 16.2 | 8.4  | 2.5                   | 100  | 0.635 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 14.9                   | 82.6            | 67.0 | 79.2 | 65.9 | 76.0 | 64.5 | 69.6                | 78.5 | 67.7 | 75.9 | 65.8 | 74.2 | 6.7                  | 50   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 66.5                            | 97.7 | 74.4 | 64.4 | 90.9 | 72.6 | 62.4 | 84.6 | 70.7 | 33.7          | 78.7 | 32.2 | 76.3 | 30.7 | 74.6 | 78.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 16.6          | 14.6 | 13.0 | -40.7                      | 88.2 | 7.5                | 3.3 | -46.1                                              | 90.5 | -50.5      | 92.5 | -54.7      | 94.3 | -60.1      | 96.7 |    |    |
| (5)               |      |      | -40.5                      | 73.6 | 7.6                | 2.1 | -46.0                                              | 75.1 | -50.4      | 76.4 | -54.7      | 77.6 | -60.2      | 79.1 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 31.2   | -3.3  | 1.2   | 18.1  | 31.6  | 46.3  | 59.8  | 64.5  | 58.2  | 46.9  | 33.3  | 13.4  | 2.4   |
|                                                                        | DBStd       | 26.16  | 16.09 | 15.21 | 11.70 | 10.96 | 10.46 | 9.06  | 7.34  | 7.32  | 7.81  | 9.35  | 14.88 | 16.25 |
|                                                                        | HDD50       | 8044   | 1653  | 1365  | 988   | 557   | 200   | 21    | 1     | 10    | 151   | 522   | 1099  | 1477  |
|                                                                        | HDD65       | 12504  | 2118  | 1785  | 1453  | 1003  | 585   | 205   | 102   | 234   | 544   | 984   | 1549  | 1942  |
|                                                                        | CDD50       | 1182   | 0     | 0     | 0     | 4     | 86    | 315   | 451   | 266   | 57    | 3     | 0     | 0     |
|                                                                        | CDD65       | 167    | 0     | 0     | 0     | 0     | 6     | 48    | 88    | 24    | 1     | 0     | 0     | 0     |
|                                                                        | CDH74       | 1147   | 0     | 0     | 0     | 0     | 58    | 340   | 590   | 155   | 4     | 0     | 0     | 0     |
|                                                                        | CDH80       | 228    | 0     | 0     | 0     | 0     | 9     | 72    | 118   | 29    | 0     | 0     | 0     | 0     |
|                                                                        | WSAvg       | 5.8    | 5.5   | 5.4   | 6.7   | 6.7   | 6.4   | 5.8   | 4.8   | 4.8   | 5.3   | 6.3   | 6.2   | 5.8   |
| (15) Precipitation                                                     | PrecAvg     | 21.2   | 1.1   | 0.9   | 0.9   | 1.2   | 1.7   | 2.5   | 3.0   | 2.9   | 2.3   | 2.0   | 1.6   | 1.3   |
|                                                                        | PrecMax     | 29.8   | 2.7   | 2.0   | 3.2   | 2.5   | 4.3   | 6.7   | 9.2   | 8.4   | 6.2   | 4.5   | 3.1   | 2.2   |
|                                                                        | PrecMin     | 13.9   | 0.2   | 0.2   | 0.0   | 0.1   | 0.2   | 0.7   | 0.6   | 0.4   | 0.6   | 0.6   | 0.4   | 0.2   |
|                                                                        | PrecStd     | 3.7    | 0.5   | 0.4   | 0.6   | 0.6   | 0.9   | 1.1   | 2.0   | 1.4   | 1.2   | 0.9   | 0.7   | 0.5   |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |
| (20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 32.5  | 34.0  | 42.9  | 64.3  | 80.7  | 86.2  | 86.9  | 84.0  | 73.6  | 58.9  | 40.9  | 33.7  |
|                                                                        |             | MCWB   | 31.6  | 32.0  | 36.4  | 53.0  | 60.0  | 66.8  | 69.9  | 70.1  | 62.1  | 51.0  | 39.5  | 32.5  |
|                                                                        | 2%          | DB     | 27.1  | 29.9  | 38.3  | 54.4  | 74.0  | 81.9  | 83.2  | 78.4  | 67.7  | 53.6  | 36.7  | 30.5  |
|                                                                        |             | MCWB   | 26.4  | 28.4  | 33.5  | 45.5  | 57.6  | 65.3  | 68.0  | 66.7  | 58.5  | 48.0  | 35.5  | 29.5  |
|                                                                        | 5%          | DB     | 22.7  | 25.4  | 35.5  | 50.0  | 68.9  | 78.0  | 80.3  | 73.7  | 62.5  | 49.8  | 34.1  | 27.2  |
|                                                                        |             | MCWB   | 22.1  | 24.2  | 32.2  | 42.3  | 55.8  | 64.3  | 67.2  | 64.4  | 55.4  | 46.0  | 32.8  | 26.4  |
|                                                                        | 10%         | DB     | 18.8  | 21.8  | 32.8  | 46.0  | 63.5  | 73.6  | 77.1  | 70.1  | 58.6  | 45.9  | 32.0  | 23.2  |
|                                                                        |             | MCWB   | 18.0  | 20.4  | 30.3  | 39.7  | 52.7  | 62.5  | 66.0  | 62.4  | 53.4  | 43.1  | 30.9  | 22.3  |
|                                                                        | 0.4%        | WB     | 31.8  | 32.6  | 36.7  | 53.2  | 62.9  | 70.7  | 72.9  | 71.1  | 63.4  | 52.2  | 39.4  | 32.4  |
|                                                                        |             | MCDB   | 32.6  | 33.9  | 41.2  | 62.8  | 74.8  | 79.5  | 81.6  | 80.7  | 71.9  | 56.3  | 40.5  | 33.2  |
|                                                                        | 2%          | WB     | 26.3  | 28.1  | 34.2  | 47.4  | 59.3  | 67.5  | 70.4  | 68.1  | 59.1  | 48.9  | 35.5  | 29.7  |
|                                                                        |             | MCDB   | 27.1  | 29.6  | 37.4  | 53.1  | 71.6  | 77.0  | 79.6  | 75.8  | 66.0  | 53.1  | 36.3  | 30.5  |
| (30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 5%          | WB     | 22.0  | 24.3  | 32.3  | 42.9  | 56.5  | 65.6  | 68.7  | 65.5  | 56.4  | 46.0  | 33.0  | 26.4  |
|                                                                        |             | MCDB   | 22.5  | 25.4  | 34.8  | 49.0  | 67.1  | 75.6  | 77.2  | 72.5  | 61.7  | 49.1  | 33.8  | 27.3  |
|                                                                        | 10%         | WB     | 18.0  | 20.6  | 30.5  | 39.9  | 53.4  | 63.6  | 67.0  | 63.4  | 53.7  | 43.0  | 30.8  | 22.3  |
|                                                                        |             | MCDB   | 18.8  | 21.9  | 32.9  | 45.4  | 63.6  | 72.7  | 75.2  | 69.3  | 58.2  | 45.4  | 31.8  | 23.1  |
|                                                                        |             | MDBR   | 12.8  | 13.3  | 15.0  | 15.3  | 16.8  | 14.9  | 14.9  | 14.0  | 12.4  | 9.4   | 10.7  | 12.9  |
|                                                                        |             | MCDBR  | 13.7  | 11.7  | 14.1  | 18.7  | 22.8  | 20.3  | 19.5  | 19.4  | 17.8  | 14.0  | 8.7   | 13.7  |
| (36) Mean Daily Temperature Range                                      | 5% DB       | MCWBR  | 13.7  | 11.3  | 11.1  | 12.2  | 12.0  | 9.8   | 8.6   | 10.0  | 10.9  | 9.4   | 8.4   | 13.3  |
|                                                                        |             | MCDBR  | 14.1  | 11.8  | 12.6  | 17.9  | 21.3  | 18.5  | 17.4  | 17.8  | 16.7  | 13.1  | 9.0   | 14.1  |
|                                                                        | 5% WB       | MCWBR  | 14.1  | 11.4  | 10.7  | 12.7  | 12.1  | 10.1  | 8.6   | 10.2  | 10.9  | 9.5   | 9.0   | 13.8  |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.257 | 0.302 | 0.354 | 0.382 | 0.395 | 0.407 | 0.421 | 0.415 | 0.358 | 0.327 | 0.273 | 0.241 |
|                                                                        | taud        |        | 1.975 | 2.010 | 1.957 | 2.058 | 2.271 | 2.330 | 2.282 | 2.257 | 2.380 | 2.272 | 2.043 | 1.914 |
|                                                                        | Ebn at Noon |        | 137   | 203   | 231   | 253   | 262   | 260   | 252   | 242   | 239   | 199   | 130   | 87    |
|                                                                        | Edh at Noon |        | 14    | 26    | 39    | 43    | 37    | 36    | 37    | 35    | 26    | 20    | 13    | 9     |
|                                                                        |             |        |       |       |       |       |       |       |       |       |       |       |       |       |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 126   | 406   | 897   | 1377  | 1596  | 1699  | 1644  | 1164  | 716   | 316   | 121   | 63    |
|                                                                        | RadStd      |        | 8     | 30    | 53    | 143   | 129   | 134   | 161   | 111   | 73    | 38    | 23    | 5     |

## Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (2 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page